

SYNAPSE · FINANCE INTELLIGENCE PLATFORM

The Story of Personalised Banking at Scale

A complete guide to how Synapse identifies each customer's needs, selects the right product, delivers the right offer, and learns from every interaction — written for anyone, regardless of banking or technology background.

PLATFORM

Synapse Finance

BANK

Meridian Bank (Demo)

CUSTOMERS

5,000

PRODUCTS

28

TRANSACTIONS

876,552

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The Problem Synapse Solves

Every bank has thousands of customers with different incomes, life stages, financial goals, and spending habits. Yet most banks send every customer the same email about the same credit card. The result: customers feel ignored, conversion rates are low, and the bank leaves revenue on the table.

Synapse fixes this by treating each customer as an individual. It reads their complete financial profile, groups them with customers who share similar traits, identifies the single product most likely to improve their financial life right now, wraps it in a personally relevant offer, and then watches how they respond — so the next recommendation is even better.

How the Platform Works — A 4-Agent Pipeline

Synapse processes every customer through four specialised agents, each handing its results to the next. Think of it as four experts in sequence: a profiler, a recommender, an offer specialist, and a session supervisor.



WHY FOUR AGENTS?

Each agent is an expert at one job. Splitting the work means each step can be tuned, audited, and improved independently — the same way a bank has separate teams for risk, product, marketing, and customer service.

Agent Architecture — Inside Each Agent

Each agent in the pipeline has a defined set of steps it runs in sequence. Here is exactly what happens inside each one, from the data it receives to what it hands off next.



AGENT C

Best Offer Selector

Input: Agent B ranked list + A/B experiment database + customer risk flags

- 1 Compliance Hard-Block**
Disqualify products that violate regulatory or risk rules regardless of score
- 2 Life Event Boost**
+20% score for products matching a recent life event (marriage, new baby, home purchase)
- 3 Churn Retention Override**
For Critical-risk customers, surface a retention offer (bonus rate, fee waiver) first
- 4 Experiment Lookup**
Match the top product to the best-performing tested offer variant (highest lift, $p < 0.05$)
- 5 Channel Attribution**
Select optimal delivery channel based on experiment attribution data per persona
- 6 Communication Preview**
Generate a personalised message tailored to the customer's preferred communication style

Output: 1 product + 1 offer variant + delivery channel + personalised message preview

AGENT D

Director & Feedback Loop

Input: Current session clicks/views + full impression ledger history

- 1 Impression Logging**
Update shown_count and click_count per customer-product pair in the ledger
- 2 Fatigue Detection**
Flag products shown ≥ 3 times with zero clicks
→ multiply score by 0.55 next session
- 3 Application Recovery**
Detect incomplete or abandoned applications
→ pin that product to position #1
- 4 Wishlist Processing**
Elevate products the customer has explicitly saved for later
- 5 Directive Writing**
Generate priority_pin, boost, and deprioritise instructions for Agent B's next run
- 6 Stack Rules Enforcement**
Final check: max 12 products, max 3 per category, no duplicate offers

Output: directives_for_next (boost/pin/deprioritise) + session behavioural summary

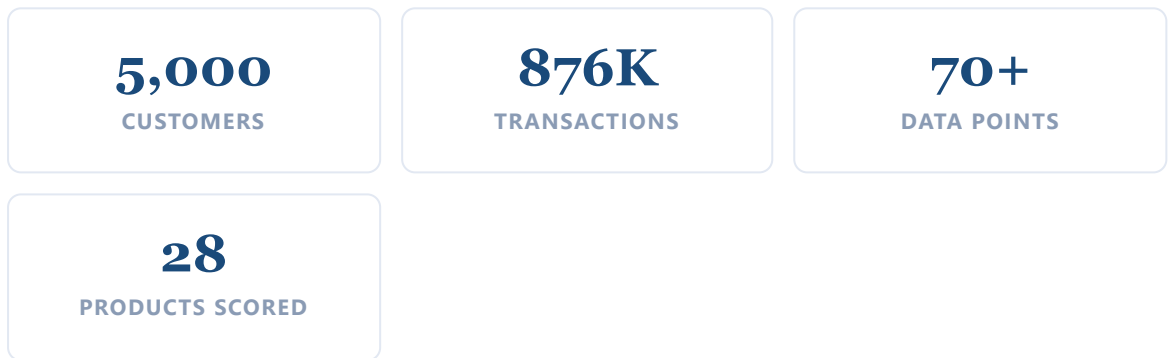
CHAPTER 1

Know Your Customer — The Dataset

1

Before making any recommendation, Synapse needs a complete picture of each customer. This chapter explains every piece of information in our customer dataset and why it matters.

The platform works with real data about 5,000 Meridian Bank customers. Each customer's record contains 70+ data points covering their finances, behaviour, preferences, and risk signals. This rich profile is what makes personalisation possible — not guesswork.



Customer Data Fields Explained

Below is every field in the customer dataset, explained in plain English.

IDENTITY & DEMOGRAPHICS

FIELD	WHAT IT MEANS
customer_id	A unique code for each customer (e.g. CUST_00001) — used to link their record across all datasets.
age	The customer's age in years — used to identify life stage and product suitability (e.g. retirement products for older customers).
income	Annual household income in USD — one of the strongest signals for which products a customer can afford and qualify for.

FIELD	WHAT IT MEANS
life_stage	A label describing where the customer is in life: Single, Young Professional, Family, Pre-Retirement, or Retired.
occupation	The customer's job title — helps understand income stability and spending patterns.
employment_type	Whether the customer is full-time, part-time, self-employed, or retired — affects income reliability.
geographic_region	Urban, Suburban, or Rural — influences which branch services are relevant.

FINANCIAL HEALTH

FIELD	WHAT IT MEANS
credit_score	A number between 300–850 showing how reliably this customer repays debts. Higher = lower risk. Primary gating criterion for loans and credit cards.
credit_tier	A label grouping the credit score: Poor, Fair, Good, or Excellent — used to match products with minimum credit requirements.
credit_score_trend	Whether the score is Rising, Stable, or Declining over the past 6 months — a declining trend is an early warning signal.
total_deposits	The total amount of money the customer holds in all their Meridian accounts right now.
outstanding_debt_total	How much the customer owes across all loans — used to avoid recommending more debt to already-stretched customers.
monthly_direct_deposit_amt	How much money lands in their account every month via payroll — a proxy for income reliability.
avg_daily_balance_p30days	The average amount in their account over the past 30 days — indicates whether they can meet minimum balance requirements.
financial_goal	What the customer is trying to achieve financially: savings growth, debt payoff, home purchase, retirement savings, or wealth building.

FIELD	WHAT IT MEANS
investment_risk_profile	How comfortable the customer is with investment risk: Conservative, Moderate, or Aggressive.
home_ownership	Whether the customer rents, has a mortgage, or owns outright — relevant for mortgage and home equity products.
fee_sensitivity_tier	How much the customer dislikes paying fees: Low, Medium, or High — prevents recommending fee-heavy products to cost-conscious customers.

BEHAVIOUR & ENGAGEMENT

FIELD	WHAT IT MEANS
digital_engagement_score	A 0–1 score of how actively the customer uses the mobile app and online banking. 1.0 = fully digital; 0.0 = never logs in.
branch_visit_frequency	How often the customer visits a physical branch — high branch usage suggests they prefer face-to-face service.
channel_preference	The customer's preferred banking channel: Digital, Branch, or Hybrid (uses both).
monthly_transaction_volume	How many transactions the customer makes each month — a measure of how actively they use their account.
txn_frequency_trend	Whether transaction activity is Increasing, Stable, or Decreasing — declining activity is a churn warning sign.
primary_spending_category	Where most of the customer's spending goes: groceries, gas, travel, dining, etc.
num_products	How many Meridian products the customer currently holds — a higher number suggests they trust the bank and are open to more.
products_held	The specific list of products currently held — used to avoid recommending something they already have.
has_direct_deposit	Whether the customer's payroll comes into their Meridian account — a strong signal of primary banking relationship.

RISK & RETENTION SIGNALS

FIELD	WHAT IT MEANS
churn_probability	A 0–1 score predicting the likelihood this customer will close their accounts or leave Meridian. Above 0.5 is high risk.
churn_risk_category	A plain-English label: Low, Medium, High, or Critical — used to trigger retention-focused offers.
overdrafts_last_3mo	Number of times the customer's account went below zero in the last 3 months — indicates financial stress.
complaints_last_year	How many formal complaints the customer raised — high numbers indicate dissatisfaction needing attention.
competitor_offer_received	Whether the customer has recently received an offer from a competing bank — an urgent signal to act.
recent_life_event	A major life change such as marriage, new baby, home purchase, or job change — prime moments to offer relevant products.

SATISFACTION & LOYALTY

FIELD	WHAT IT MEANS
nps_score	Net Promoter Score (0–100) — how likely the customer is to recommend Meridian. Below 50 = Detractor; 70+ = Promoter.
customer_lifetime_value	The projected total revenue Meridian expects to earn from this customer over their entire relationship, in USD.
clv_tier	A value tier label: Bronze, Silver, Gold, or Platinum — determines priority level of service and offers.
cross_sell_receptivity	A 0–1 score measuring how open this customer has been to adopting new products in the past.
preferred_rewards_type	Whether the customer prefers Cash Back, Points, or Travel Miles — used to match the right credit card rewards.
preferred_communication_style	How the customer likes to receive information: through numbers and data, or through stories and relationships.

Key Metric Formulas

NET PROMOTER SCORE (NPS)

FORMULA

NPS = % Promoters (score 9–10) – % Detractors (score 0–6)

Range: -100 to +100 | Passive (7–8) counts in neither group

Example: 45% Promoters, 20% Detractors → NPS = 25. Above 50 is considered strong for retail banking.

CUSTOMER LIFETIME VALUE (CLV)

FORMULA

CLV = Average Annual Revenue per Customer
× Projected Relationship Length (years)
× (1 – Churn Probability)

A customer generating \$800/yr with a 12-year expected tenure and 10% churn probability: CLV ≈ \$8,640.

CHURN PROBABILITY

FORMULA (LOGISTIC MODEL)

Churn_Prob = $1 / (1 + e^{-z})$

where $z = \beta_0 + \beta_1 \cdot \text{balance_decline} + \beta_2 \cdot \text{txn_decline}$
 $+ \beta_3 \cdot \text{complaints} + \beta_4 \cdot \text{competitor_signal} + \dots$

Range: 0.0 (no risk) to 1.0 (certain churn)

A falling balance AND falling transactions AND a competitor offer received pushes z strongly negative → high churn probability.

CHAPTER 2

The Product Catalogue

Meridian Bank offers 28 financial products. Synapse knows every detail about each one — costs, requirements, rates, and rewards — so it can match the right product to the right customer.

The product catalogue is the menu from which Synapse selects recommendations. Every product has a set of eligibility rules (minimum credit score, minimum income) and a set of value signals (interest rate, fees, rewards) that help the platform decide who would benefit most from it.

PRODUCT CATEGORIES

CATEGORY	WHAT IT IS	EXAMPLES
Checking Account	Everyday spending account with debit card, Zelle, and mobile deposit	Essential Checking, Plus Checking
Savings Account	Account that earns interest on money set aside	High-Yield Savings, Money Market
Certificate of Deposit	Fixed-term deposit at a locked-in interest rate — higher rate in exchange for not withdrawing early	12-Month CD, 24-Month CD
Credit Card	Revolving credit line for purchases; may have rewards, cashback, or travel miles	Cashback Rewards Card, Travel Elite Card
Loan	Fixed-amount borrowing repaid over time with interest	Personal Loan, Auto Loan
Mortgage	Long-term loan secured against a property	30-Year Fixed Mortgage
Investment / Wealth	Products that grow money over time through markets or managed portfolios	Investment Account, IRA

Product Data Fields Explained

FIELD	WHAT IT MEANS
product_id	Unique identifier for the product (e.g. MRD_P001).
product_name	The name customers see, e.g. "Meridian Essential Checking".
annual_fee	The flat fee charged once per year to hold this product. Zero for most basic accounts.
monthly_fee	A recurring monthly charge — often waivable by meeting certain conditions (e.g. direct deposit).
min_balance	The minimum amount of money required in the account to avoid fees or maintain the account.
interest_rate / APY	The annual return on savings or CD products. See APY formula below.
regular_apr_pct	Annual Percentage Rate on loans and credit cards — the yearly cost of borrowing money.
intro_apr_months	Number of months a promotional 0% APR applies before the regular rate kicks in.
min_credit_score	The lowest credit score a customer needs to qualify for this product.
credit_tier_required	Hard eligibility gate — some premium products require "Excellent" credit tier regardless of score.
min_income	Minimum annual income required to qualify.
rewards_type	Type of rewards offered: Cash Back, Points, Travel Miles, or None.
cash_back_pct	Percentage of spend returned as cash — e.g. 2% means \$2 back for every \$100 spent.
fee_waiver_condition	The rule that removes the monthly fee — e.g. "set up \$500+ direct deposit".
digital_required	True/False — some products are online-only and unavailable to branch-only customers.
complexity_score	How complicated the product is to use or understand: 1 = simple, 2 = moderate, 3 = complex.

FIELD	WHAT IT MEANS
avg_rating	Average customer satisfaction rating out of 5 stars, based on reviews and surveys.
aum_fee_pct	Assets Under Management fee — a percentage of the invested balance charged annually on wealth products.

What is APY?

PLAIN ENGLISH DEFINITION

APY (Annual Percentage Yield) is the real annual return on a savings account or CD, including the effect of compound interest. A savings account advertising 4.5% APY means if you put \$10,000 in it, you'll have approximately \$10,450 at the end of the year — without doing anything.

APY FORMULA

$$APY = (1 + r / n)^n - 1$$

where r = nominal annual interest rate (e.g. 0.044 for 4.4%)
 n = number of compounding periods per year
(daily = 365, monthly = 12, quarterly = 4)

Example: 4.4% rate compounded daily $\rightarrow APY = (1 + 0.044/365)^{365} - 1 \approx 4.50\%$. APY is always $\geq$ the nominal rate. Synapse uses APY so all savings products are compared on the same footing.

WHY SYNAPSE USES APY IN SCORING

When comparing savings products for a customer, Synapse uses APY to fairly compare products that compound at different frequencies. A customer with \$50,000 in savings gets substantially more from a 4.5% APY product than a 3.8% APY product — Synapse quantifies this and factors it into the recommendation score.

CHAPTER 3

Segmenting the Customer Base — Agent A

Not all customers are alike. Agent A's job is to group 5,000 unique individuals into meaningful segments so that every downstream recommendation is anchored in who the customer really is.

Imagine trying to recommend a product to every one of 5,000 customers individually — there simply aren't enough human advisors. But recommending the same thing to everyone wastes the opportunity. Agent A solves this by finding natural groups within the customer base: people who share similar financial profiles, behaviours, and goals.

The algorithm reads every customer's income, credit score, deposits, transaction patterns, digital engagement, and risk signals simultaneously, then finds five natural clusters.

THE ALGORITHM: K-PROTOTYPES

Synapse uses an algorithm called **K-Prototypes** — a method that handles both numbers (like income and credit score) and categories (like "Urban" or "Hybrid channel preference") simultaneously. Most clustering algorithms only work with numbers. K-Prototypes is specifically designed for real-world datasets that mix both, making it ideal for banking customer data.

The Five Customer Personas

After clustering, each group is given a name that captures their essence. These names are generated by AI and grounded in the real data characteristics of that cluster.

Wealth Builder**HIGH INCOME · INVESTMENT-FOCUSED**

High earners (income $\geq$ \$128K) actively growing their net worth. Most receptive to investment, premium credit, and wealth management products.

Everyday Essential**MASS MARKET · CORE BANKING**

The core of any bank's customer base — steady income, moderate balances, focused on practical products like savings accounts and basic loans.

Digital Native**YOUNG · MOBILE-FIRST**

Young, tech-comfortable customers who rarely visit branches. Digital engagement score above 0.79. Respond best to in-app offers and instant account opening.

Retiree Anchor**AGE 63+ · BRANCH PREFERRED**

Long-standing customers who prefer branch banking. Lower digital engagement but strong loyalty. Most suited for CDs, savings, and insurance products.

At-Risk Saver**HIGH CHURN RISK · FEE-SENSITIVE**

Customers showing early warning signs: declining balances, missed payments, or competitor offers received. Require retention-focused, low-fee product recommendations.

How Good is the Segmentation? The Silhouette Score

SILHOUETTE SCORE — PLAIN ENGLISH

Think of it as a quality rating for the grouping. It asks two questions about every customer: "*How similar are you to others in your group?*" and "*How different are you from the nearest other group?*" The answer is a number between -1 and +1. Above 0.5 means the groups are well-defined. Below 0.2 means the groups are fuzzy and overlapping. Synapse targets a score above 0.5.

SILHOUETTE SCORE FORMULA

$$s(i) = (b(i) - a(i)) / \max(a(i), b(i))$$

where $a(i)$ = mean distance from customer i to all others in its cluster
 $b(i)$ = mean distance from customer i to all customers
in the nearest other cluster

Final score = mean $s(i)$ across all 5,000 customers

Range: -1 (wrong cluster) → 0 (on boundary) → +1 (perfectly placed)

Synapse runs the score after each Agent A execution. If it falls below 0.45, the run is flagged for review — the five groups may not be meaningfully distinct.

Macro Segments vs. Micro Segments

	MACRO SEGMENT	MICRO SEGMENT
What it is	One of the 5 broad customer personas (e.g. Wealth Builder)	A sub-group within a persona, carved by additional traits
Example	"Everyday Essential" — 1,200 customers	"Everyday Essential · Urban · Age 30–45 · Direct Deposit" — 180 customers
Used for	Campaign targeting, persona-level marketing strategy	Hyper-targeted product offers, micro-segment ROAS analysis
How created	Primary K-Prototypes clustering on all 5,000 customers	Secondary clustering run within each macro group using finer attributes
Value	Broad strategic insight — "who are our customer types?"	Precision targeting — "which sub-group within Digital Natives is ready for a credit card right now?"

BUSINESS VALUE

A macro segment tells you the customer type. A micro segment tells you the exact moment and message. Together, they let Meridian Bank think like a boutique private bank — even at 5,000-customer scale.

CHAPTER 4

The Recommendation Engine — Agent B & the 12 Factors

Agent B scores every one of the 28 products for every customer using 12 carefully chosen factors. The result is a personalised ranking — the right product at the top, for the right reason.

For each customer, Agent B runs through all 28 products and asks: "How good is this product for this specific person?" It scores each combination using 12 factors, each weighted by how much it should influence the final recommendation.

COMPOSITE RECOMMENDATION SCORE

```
Score(customer, product) =  $\sum_{i=1}^{12} \text{Factor\_Score}_i \times \text{Weight}_i$  for i = 1 ... 12
```

```
Final_Score = (raw_score / max_possible_score) × 100
```

Each `Factor_Score_i` is normalised 0-1 before weighting.

Weights sum to 1.0 (100%). Products ranked descending by `Final_Score`.

336 individual scores are computed per customer (28 products × 12 factors). The top 12 non-blocked results form the recommendation stack.

The 12 Factors — What They Are and Why They Matter

Factor 01

22%

Credit & Income Eligibility

Checks whether the customer actually qualifies — credit score, income, and deposit balance all meet minimum thresholds. The highest-weighted factor because recommending a product the customer can't get is worse than no recommendation at all.

```
score = min(credit/min_credit, 1) ×  
0.50 + min(income/min_income, 1) × 0.30  
+ min(balance/min_balance, 1) × 0.20
```

Factor 02

18%

Financial Need Match

Does this product align with what the customer is trying to achieve? A customer with a "home purchase" goal scores higher on mortgages. Ensures recommendations feel relevant, not random.

```
score =  
goal_match_weight[customer.goal]  
[product.type] × (1 -  
abs(income_centile - product_centile))
```

Factor 03

13%

Relationship Depth

Rewards loyalty. Customers with longer tenure and more existing products score higher — cross-selling to engaged customers converts better than cold-selling.

```
score = tenure_norm × 0.60 +  
(num_products / 10) × 0.40
```

Factor 04

13%

Product Financial Value

How much is this product actually worth to the customer? Combines fee cost, interest rate benefit, and projected return. A 4.5% APY savings account scores much higher than a 1.2% one.

```
score = (APY_benefit - annual_fee /  
balance) normalised across all products  
in category
```

Factor 05

10%

Product Pairing

Some products work better together. A customer with a checking account scores higher on savings accounts (natural pairing). Captures cross-sell logic that an experienced banker would know intuitively.

```
score = pairing_affinity[products_held  
→ product] (lookup table, 0.0 - 1.0)
```

Factor 06

6%

Intent Score

A composite measure of how ready this customer is to say yes right now. Combines behavioural signals and preference signals. See full formula in this chapter.

```
score = Intent_Score range 0.08 - 0.97
```

Factor 07

4%

Channel Fit

Some products are digital-only. This factor ensures those products are only recommended to customers comfortable banking digitally. Avoids recommending an app-only product to a customer who never logs in.

```
score = 1.0 if channel_pref matches  
product.channel 0.5 if Hybrid 0.1 if  
mismatch
```

Factor 08

4%

Complexity Match

Aligns product sophistication with customer financial knowledge — protecting customers from products they might misuse or regret.

```
score = 1 - abs(complexity_score -  
sophistication) / 2 (normalised 0-1)
```

Factor 09

4%

Product Rating

Real customer satisfaction scores (out of 5 stars). A product that existing customers love is more likely to satisfy the next customer too.

```
score = (avg_rating - 1) / 4 (maps 1-5  
stars to 0.0-1.0)
```

Factor 10

3%

Popularity in Segment

How many other customers in the same persona currently hold this product. Social proof — if 60% of Digital Natives hold this card, the next Digital Native is likely to want it too.

```
score = count(segment_holders) /  
segment_size
```

Factor 11

2%

Segment Affinity

Historical adoption rate of the product within the customer's micro-segment. A more targeted measure than raw popularity.

```
score = adopted_in_micro_segment /  
total_in_micro_segment
```

Factor 12

1%

Transaction Pattern

The volume and frequency of the customer's recent transactions. Highly active customers are more engaged with their finances and more likely to appreciate a new product.

```
score = min(monthly_txn_volume / 30,  
1.0) (capped at 30 txns/month = 1.0)
```

WHY 12 FACTORS?

A single factor — say, credit score — would get the eligibility right but miss everything else. A customer might qualify for ten products but only actually need one. The 12 factors together capture eligibility, desire, timing, value, behaviour, and social signals simultaneously. Each factor catches something the others miss, reducing the chance of a recommendation that is technically correct but contextually wrong.

Intent Score — Is This Customer Ready to Buy?

The Intent Score is a single number (0 to 0.97) that answers: *How likely is this customer to say yes to a new product right now?*

HOW IT'S BUILT

Two signals are combined. The **Behaviour Signal** (75% weight): how active they are — transaction frequency, how fast they adopt new products, and their credit health. The **Preference Signal** (25% weight): their stated willingness (cross-sell receptivity) and how digitally engaged they are. The final score is capped at 0.97 — no one is a guaranteed yes.

INTENT SCORE FORMULA

```

Behaviour_Signal = txn_frequency_norm × 0.40
                  + adoption_speed_norm  × 0.35
                  + credit_health_norm    × 0.25

Preference_Signal = cross_sell_receptivity × 0.60
                  + digital_engagement    × 0.40

Intent_Score = clip(
    Behaviour_Signal × 0.75 + Preference_Signal × 0.25,
    min = 0.08, max = 0.97
)

```

clip() ensures the score never reaches 0 (everyone has some potential) or 1.0 (no one is a guaranteed conversion). A score ≥ 0.70 is treated as high-intent; ≤ 0.30 is low-intent.

Digital Adoption Score

DIGITAL ADOPTION SCORE FORMULA

```

Digital_Adoption = digital_engagement_score × 0.70
                  + (1 - branch_visit_norm) × 0.30

```

where `digital_engagement_score` is already 0-1 from the dataset

```
branch_visit_norm = branch_visit_frequency / max_branch_freq
```

Inverting branch visits means high branch attendance pulls the score down. This score gates digital-only product recommendations. Above 0.65 = highly digital. Below 0.40 = branch-preferring.

CHAPTER 5

The One Best Offer — Agent C

From Agent B's ranked list, Agent C selects a single product and wraps it in the offer most likely to convert — backed by real A/B experiment results, not hunches.



Showing a customer ten products is as bad as showing them none — the choice is overwhelming. Agent C's job is to make one clear, confident decision: this customer, this product, this offer, delivered this way.

The selection process checks eligibility compliance (hard rules), applies life-event boosts, and applies churn retention logic before querying the experiment database for the best-tested offer variant.

COMPLIANCE HARD BLOCKS

RULE	WHY IT EXISTS
Credit tier "Excellent" required — customer is "Good"	Prevents the embarrassment and regulatory risk of offering a premium product a customer will be rejected for.
Annual product cost exceeds 1% of income AND customer is fee-sensitive	Protects financially stressed customers from products that would strain their budget.
High-APR rewards card when customer has \$20K+ outstanding debt	Avoids recommending expensive revolving credit to customers already carrying heavy debt loads.

RULE	WHY IT EXISTS
Customer has already churned	No credit or investment offers for customers who have already closed accounts — a retention path is needed first.

Lift & Holdout Rate — Did Our Offer Actually Work?

Every offer in Synapse is backed by a real experiment that tested whether the offer made a measurable difference. The two key metrics from these experiments are **Lift** and **Holdout Rate**.

HOLDOUT RATE — PLAIN ENGLISH

The holdout group is the **control group** — a set of customers who were shown *no offer at all*. They represent the natural baseline: how many people would have signed up for this product on their own, without any nudge from us.

HOLDOUT RATE FORMULA

$$\text{Holdout_Rate} = \frac{\text{Conversions_in_control_group}}{\text{Total_customers_in_control_group}} \times 100$$

Example: 56 sign-ups out of 400 control customers

$$\rightarrow \text{Holdout Rate} = 56 / 400 \times 100 = 14\%$$

LIFT — PLAIN ENGLISH

Lift is the extra conversions we generated because of our offer. If the holdout group converts at 14% and our offer group converts at 23%, the lift is 9 percentage points — meaning our offer caused 9 additional customers out of every 100 to sign up who otherwise would not have.

LIFT FORMULAS

Lift_pp = Conversion_Rate_Treatment - Conversion_Rate_Holdout
(percentage point lift – the raw difference)

Lift_pct = (Conversion_Rate_Treatment - Conversion_Rate_Holdout)
/ Conversion_Rate_Holdout × 100
(relative percentage uplift)

Example: (23% - 14%) / 14% × 100 = 64% lift
→ Our offer made conversions 64% better than doing nothing.

STATISTICAL SIGNIFICANCE — IS THE RESULT REAL?

Z-TEST FOR STATISTICAL SIGNIFICANCE

z = (p_treatment - p_holdout)
/ √[p_pooled × (1 - p_pooled) × (1/n_t + 1/n_h)]

where p_treatment = conversion rate of offer group
p_holdout = conversion rate of control group
p_pooled = combined conversion rate of both groups
n_t, n_h = sample sizes of each group

p-value < 0.05 → 95% confident the result is real, not luck
p-value < 0.01 → 99% confident (Synapse flags this as "strong signal")

Only statistically significant offers (p < 0.05) are stored in the experiment database and available for Agent C to select.

A Real Experiment — Digital Native + Checking Bonus

EXPERIMENT: SUMMER \$300 CHECKING BONUS · DIGITAL NATIVE PERSONA · IN-APP CHANNEL				
GROUP	CUSTOMERS SHOWN	SIGNED UP	CONVERSION RATE	RESULT
Holdout (no offer)	400	56	14%	Baseline
Variant A (\$300 Bonus)	400	92	23%	+9pp lift · 64% uplift ·

				p=0.0005 ✓
--	--	--	--	-------------------

CHANNEL ATTRIBUTION

In this experiment, 87% of conversions came from the in-app channel, 4.3% from email, 4.3% from direct mail, and 4.3% from branch. This tells Agent C not just *which offer to make*, but *where to deliver it* — in-app for Digital Natives, branch for Retiree Anchors, email for Everyday Essentials.

CHAPTER 6

The Director Agent — Agent D



Agent D watches what happens after a recommendation is made. It tracks which products customers see, which they engage with, and uses that signal to make the next visit smarter.

A recommendation engine that never learns from experience is just a static list generator. Agent D closes the loop — it observes every customer session, logs every product shown and every click, and issues instructions to Agent B for the next time that customer visits.

The Impression Ledger

WHAT IS AN IMPRESSION?

An impression is one instance of a product appearing in a customer's recommendation stack. Every time a customer sees "Meridian Cashback Card" in their recommendations, that's one impression. If they click on it, that's a click. The ledger is a running record of impressions and clicks for every customer-product combination.

METRIC	WHAT IT TRACKS	HOW IT'S USED
shown_count	How many times this product has been shown to this customer across sessions	If shown 3+ times with zero clicks, the product is deprioritised next session
click_count	How many times the customer engaged with this product	Clicked products get boosted to the top of the next session's stack
last_shown_date	When the product was last recommended	Ensures product rotation — prevents the same offer appearing every single session

ENGAGEMENT RATE PER PRODUCT (FROM IMPRESSION LEDGER)

Engagement_Rate = `click_count` / `shown_count`

< 0.10 → Deprioritise: apply 0.55× score multiplier next session

≥ 0.10 → Maintain or boost

≥ 0.30 → High interest: apply `priority_pin` directive

A `shown_count` of 3 with 0 clicks gives engagement rate = 0.0 — this triggers the fatigue penalty. A single click after 2 shows gives 0.50 — this triggers a boost.

Directives — Instructions for the Next Visit

DIRECTIVE	WHAT IT DOES	EXAMPLE
Priority Pin	Forces one product to position #1 in the next session	Customer started a credit card application but didn't finish — pin that card to #1 next visit
Boost Products	Adds +12% to the score of products the customer showed interest in	Customer clicked on "Meridian High-Yield Savings" — boost savings products next session
Deprioritise Products	Reduces score by 45% for products shown repeatedly with no engagement	Customer has seen the auto loan 4 times and never clicked — deprioritise it
Focus Source	Tells Agent B where the priority signal is coming from	"incomplete_application" means the customer has an abandoned application to recover

PRIORITY ORDER (HIGHEST TO LOWEST)

1. A product the customer clicked in their most recent session
2. An incomplete application they started but abandoned
3. A product saved to their wishlist
4. Agent C's commercially selected best offer
5. Agent B's highest-scoring product (fallback)

LIVE STACK RULES

RULE	WHY IT EXISTS
Maximum 12 products in the live recommendation stack	Beyond 12, customers experience choice overload and convert less
Maximum 3 products per category (e.g. max 3 credit cards)	Prevents category flooding — a stack full of credit cards feels like an upsell, not a service

CHAPTER 7

Ad Spend Intelligence — Channel ROI Dashboard

Knowing which products to recommend is only half the battle. Synapse also tells marketers exactly how much to spend on each channel, and whether they're getting a return on that spend.

The Ad Spend module analyses 12 months of marketing data across four channels (email, social, direct mail, and branch) and five customer personas. It tells the marketing team not just what worked, but what they should do differently next month.

Channel Metrics — What Each Number Means & How It's Calculated

METRIC	PLAIN ENGLISH MEANING	FORMULA	GOOD SIGNAL
ROAS	For every \$1 spent on marketing, how many \$1s of product revenue came back?	$\text{Revenue} / \text{Ad_Spend}$	Above 1.0 = profitable

METRIC	PLAIN ENGLISH MEANING	FORMULA	GOOD SIGNAL
True ROAS	Like ROAS but adjusts for operating costs — a more honest view of actual profit per dollar spent.	$(\text{Revenue} - \text{Op_Cost}) / \text{Ad_Spend}$	Higher than ROAS target after costs
mROAS	The return on the <i>next</i> dollar spent on this channel right now.	$\Delta \text{Revenue} / \Delta \text{Spend}$	Above 1.25 = invest more; below 0.80 = cut back
ROI %	Net profit percentage of the marketing investment.	$(\text{Revenue} - \text{Spend}) / \text{Spend} \times 100$	Positive = profitable campaign
CPA	How much was spent in marketing for every one customer who converted.	$\text{Ad_Spend} / \text{Conversions}$	Lower is better vs customer CLV

METRIC	PLAIN ENGLISH MEANING	FORMULA	GOOD SIGNAL
Conversion Rate	Percentage of people who received the message and then signed up.	$\text{Conversions} / \text{Contacted} \times 100$	Above 20% is strong in banking
Open Rate	Percentage of people who opened the email or direct mail piece.	$\text{Opened} / \text{Sent} \times 100$	25–40% is strong for banking email
Click Rate	Percentage of people who clicked a link or call-to-action in the marketing piece.	$\text{Clicked} / \text{Opened} \times 100$	Above 10% is excellent
Saturation %	How close a channel is to its maximum effective spend. At 100% saturation, additional spend returns nothing new.	$\text{Current_Spend} / \text{Max_Effective_Spend} \times 100$	Under 70% = room to grow

MROAS — UNDERSTANDING THE MARGINAL RETURN

MARGINAL ROAS (MROAS) DETAIL

$$mROAS = (\text{Revenue}(\text{spend} + \Delta) - \text{Revenue}(\text{spend})) / \Delta$$

where Δ = a small incremental spend increase (e.g. \$1,000)

Derived from the S-Curve model:

$$\text{Revenue} = a \times (1 - e^{(-b \times \text{spend})})$$

$$mROAS = a \times b \times e^{(-b \times \text{spend})}$$

(declines as spend increases — diminishing returns)

As a channel approaches saturation, mROAS falls. When mROAS for channel A < mROAS for channel B, reallocate budget from A to B until they equalise — this is the optimal allocation point.

S-CURVE IN PLAIN ENGLISH

When you first start spending on a channel, each dollar works very hard — you're reaching receptive customers who haven't seen the message yet. As you spend more, returns grow quickly. But at some point, you've reached everyone who's likely to respond — and the last few dollars are going to people who will never convert. The S-Curve shows this journey: slow start, rapid middle growth, then a flattening plateau. Synapse maps exactly where each channel sits on its S-Curve so you know whether to accelerate or redirect spend.

CONVERSION FUNNEL — WHERE ARE WE LOSING CUSTOMERS?

FUNNEL STAGE	WHAT IT MEASURES	FORMULA	DROP-OFF MEANING
Sent	Total contacts reached with the marketing message	—	—
Opened	How many opened or viewed the message	$\text{Opened} / \text{Sent} \times 100$	Low = wrong subject line, wrong timing, or wrong segment
Clicked	How many engaged with the call-to-action	$\text{Clicked} / \text{Opened} \times 100$	Low = weak offer or wrong product for this audience

FUNNEL STAGE	WHAT IT MEASURES	FORMULA	DROP-OFF MEANING
Converted	How many actually signed up for the product	$Converted / Clicked \times 100$	Low despite high clicks = friction in the application process itself

CHAPTER 8

Segment AI — Conversational Analytics

A built-in AI chatbot that lets anyone on the team ask plain-English questions about customers, segments, and performance — and get answers backed by live data.

Not every business question belongs in a dashboard. Sometimes you need to ask something specific: *"Which customers in the Wealth Builder segment have more than \$100K in deposits but no investment account?"* Segment AI makes this possible without needing a data analyst or SQL knowledge.

Under the hood, the chatbot connects to all of Meridian Bank's data stored in Databricks Unity Catalog — a secure, governed data platform. When you type a question, it generates a precise database query, runs it live, and returns the answer alongside the data table and a chart.

WHAT MAKES IT TRUSTWORTHY

Every answer shows the SQL query it used — so any analyst can verify the logic. The data comes directly from your Unity Catalog Delta tables — not cached, not approximated. If the question can't be answered with the available data, the system says so and suggests a reformulation.

WHAT YOU CAN ASK

QUESTION TYPE	EXAMPLE
Churn risk analysis	"Which segment has the highest churn risk and what is the total revenue at risk?"
Customer profiling	"Show me the top 10 customers by lifetime value who hold no investment products"
Segment comparison	"Compare average income, credit score, and transaction volume across all five segments"
Marketing performance	"Which channel gives the best return for the Digital Native persona?"
Opportunity identification	"Which customers have incomplete applications we should follow up with?"

QUESTION TYPE	EXAMPLE
Retention targeting	"List Critical churn-risk customers who are Gold or Platinum CLV tier"

REFERENCE

Glossary — Every Term Explained

A quick-reference dictionary of every metric, score, and concept used in Synapse Finance.



APY (Annual Percentage Yield)

The real annual return on a savings or CD account, including compound interest.
Formula: $(1 + r/n)^n - 1$.

APR (Annual Percentage Rate)

The yearly cost of borrowing money on a loan or credit card. Lower APR = cheaper to borrow.

CLV (Customer Lifetime Value)

Projected total revenue from a customer: Avg Annual Revenue $\times$ Tenure Years $\times$ (1 – Churn Prob).

CLV Tier

A value label: Bronze $\rightarrow$ Silver $\rightarrow$ Gold $\rightarrow$ Platinum. Determines service priority and offer quality.

Churn Probability

A 0–1 logistic score: $1/(1+e^{-z})$ where z encodes balance decline, transaction decline, complaints, and competitor signals. Above 0.5 = high risk.

CSAT

Customer Satisfaction Score (1–5) from recent service interactions. Used in product scoring.

CPA (Cost Per Acquisition)

Ad_Spend / Conversions. How much marketing spend was needed to convert one customer.

Cross-Sell Receptivity

A 0–1 score of how open this customer has been to adopting new products historically. Key input to Intent Score.

Digital Adoption Score

$\text{digital_engagement} \times 0.70 + (1 - \text{branch_norm}) \times 0.30$. Gates digital-only product recommendations.

Directive

A written instruction from Agent D to Agent B about how to prioritise the next session's recommendations.

Holdout Rate

$\text{Conversions} / \text{Total_Control} \times 100$. The baseline conversion rate of customers who received no offer.

Impression

One instance of a product appearing in a customer's recommendation stack in a session.

Intent Score

K-Prototypes

$\text{Behaviour_Signal} \times 0.75 + \text{Preference_Signal} \times 0.25$, clipped to [0.08, 0.97]. Measures readiness to adopt a product.

The clustering algorithm Agent A uses. Handles both numeric and categorical data simultaneously.

Lift (pp)

$\text{Conversion_Rate_Treatment} - \text{Conversion_Rate_Holdout}$. The extra sign-ups generated by the offer.

Lift (%)

$(\text{Treatment} - \text{Holdout}) / \text{Holdout} \times 100$. Relative percentage improvement over doing nothing.

Macro Segment

One of the 5 broad customer personas (Wealth Builder, Digital Native, etc.) — the top-level grouping.

Micro Segment

A sub-group within a macro segment, defined by additional attributes for hyper-targeted campaigns.

mROAS (Marginal ROAS)

$\Delta \text{Revenue} / \Delta \text{Spend}$. The return on the next dollar spent. Above 1.25 = invest more; below 0.80 = cut back.

NPS (Net Promoter Score)

$\% \text{Promoters (9–10)} - \% \text{Detractors (0–6)}$. Range –100 to +100. Above 50 is strong in retail banking.

Priority Pin

A directive that forces one specific product to the #1 position in the customer's next recommendation session.

ROAS (Return on Ad Spend)

$\text{Revenue} / \text{Ad_Spend}$. ROAS of 3.0 means \$3 back for every \$1 spent.

Saturation

$\text{Current_Spend} / \text{Max_Effective_Spend} \times 100$. Near 100% means additional budget generates no new revenue.

Silhouette Score

$(b(i) - a(i)) / \max(a(i), b(i))$. Quality rating for segmentation. Above 0.5 = well-defined clusters.

True ROAS

$(\text{Revenue} - \text{Operating_Cost}) / \text{Ad_Spend}$. More conservative and accurate than raw ROAS.

Unity Catalog

The secure, governed Databricks data platform where all Meridian Bank tables live and are queried by Segment AI.

Z-Score (Statistical Test)

$(p_{\text{treatment}} - p_{\text{holdout}}) / \text{SE}$. $|z| > 1.96 \rightarrow p < 0.05 \rightarrow$ result is statistically significant.

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